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Stel de rij is $a, a + d, a + 2d, \dots$

$$\Rightarrow a + d = 3(a + 2d)$$

$$\Rightarrow a + d = 3a + 6d$$

$$\Rightarrow -2a - 5d = 0$$

$$\Rightarrow 2a = -5d$$

$$\Rightarrow a = -\frac{5}{2}d$$

Als $d = 0 \Rightarrow a = 0 \Rightarrow$ nulrij

Als $d \neq 0$:

$$S_n = \frac{a + a + (n-1)d}{2} \cdot n = 0$$

$$\Rightarrow \frac{2a + (n-1)d}{2} \cdot n = 0$$

$$\Rightarrow 2a + (n-1)d = 0$$

$$\Rightarrow -5d + (n-1)d = 0$$

$$\Rightarrow (-5 + n - 1)d = 0$$

$$\Rightarrow (n - 6)d = 0$$

$$\Rightarrow n = 6$$

References